

Some Tips **BEYOND:** Limits

0- We describe a function value at a point as:

$f(a)$ is **defined** if $a \in D_f$ **OR** $f(a)$ is **undefined** if $a \notin D_f$.

1- The limit of a function " $\lim_{x \rightarrow a} f(x)$ " is a **y-value**. Limits are described as:

"Exists" (Determined), **"Undetermined"**, **OR** **"Does Not Exist (DNE)"**.

2- One may describe the "limit" concept as a mathematical tool that could be used to:

- a. Evaluate the value of a function at "nearby" a point where the function is undefined, if possible.
- b. Study the attitude of functions "applications" at **infinity** or at **singularities**, where nobody can visit:
 - i. Like measuring the temperature of a volcano or the temperature at the center of the earth or inside the sun.
 - ii. Like measuring the diameter of earth or the speed of light.
 - iii. Like measuring the distances between earth and other planets.
 - iv. etc...
- c. Redefine a well-defined function to achieve a certain task, e.g. removable discontinuities.
- d. Convert approximate values, e.g. " $e \approx 2.718$ " to exact values, e.g. " $e = \lim_{x \rightarrow \infty} (1 + \frac{1}{x})^x$ ".
- e. etc...

3- **If a limit** of a function "at a certain point" **exists**, therefore it is **unique**. That means **EXACT** value.

4- One may summarize the **TECHNIQUES** of evaluating a limit value as follows:

- a. Direct Substitution may give a **determined** value, therefore the limit **exists**.
- b. Direct Substitution may give an **undetermined value**, then the limit **may or may not exist** based on the following cases:

Undetermined values of limits

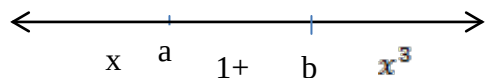
$\frac{0}{0}$	$\frac{\pm \infty}{\pm \infty}$	$0.(\pm \infty)$	$\infty - \infty$	$1^{\pm \infty}, 0^0, (\pm \infty)^0$	$\frac{\text{Number}}{0} = \pm \infty$			
From Trigonometric functions:- <u>Use:-</u> $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ & $\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$	(To cancel zero factor) 1- Factorize up & down. OR 2- Multiply by the Conjugate.	L'Hopital (standard case) $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$	(∞ term) We divide up & down by the term "in DENO" that produces $+\infty$. <u>Note:</u> [Sometimes if we have square roots, it is better to take a common factor out the root]	L'Hopital (special case1) $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$	L'Hopital (special case2) $f(x).g(x)$ <u>Answer key:</u> $\frac{f(x)}{1/g(x)}$ OR $\frac{g(x)}{1/f(x)}$	L'Hopital (special case2) $f(x) - g(x)$ <u>Answer key:</u> 1- Make common denominator. OR 2- Take common factor.	L'Hopital (special case3) $f(x)^{g(x)}$ <u>Answer key:</u> 1- $y = f^g$ 2- $\ln(\cdot)$ for both sides. 3- limit for both sides. 4- Get $e^{\text{R.H.S}}$	1- Separate the limit into two limits. 2- Evaluate the limit which has zero in the denominator using the right and left sided limits to get $\frac{1}{0}$ which comes always from $\frac{1}{x}$ or $\frac{1}{x^2}$.

One-Sided limits “right & left sided limits”

1- If the function is a piecewise function
(Defined Differently with cases)

Ex:-

$$f(x) = \begin{cases} x & x < a \\ 1+x & a < x < b \\ x^3 & x > b \end{cases}$$



$$\lim_{x \rightarrow a} f(x) = \begin{cases} \lim_{x \rightarrow a^+} 1+x & \text{from the right} \\ \lim_{x \rightarrow a^-} x & \text{from the left} \end{cases}$$

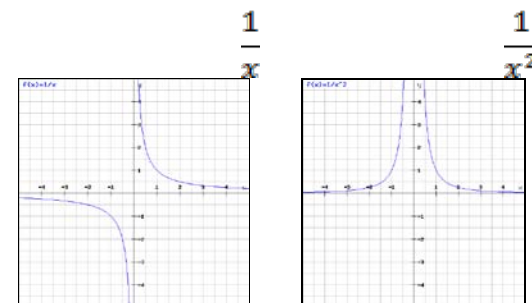
2- If the function contains a modulus such that the study point of the limit is same as the symmetric point of the absolute value term.

Ex:-

$$\lim_{x \rightarrow b} \frac{|x-b|}{x-b} = \begin{cases} \lim_{x \rightarrow b^+} \frac{x-b}{x-b} & \text{from the right} \\ \lim_{x \rightarrow b^-} \frac{-(x-b)}{x-b} & \text{from the left} \end{cases}$$

3- Direct substitution = $\frac{\text{Number}}{0}$

Which comes from either:



If a limit includes **bounded** functions, one may use the sandwich “squeezing” theorem.

Use the following:-

$$-1 \leq \sin(x) \leq 1$$

$$0 \leq |\sin(x)| \leq 1$$

$$0 \leq (\sin(x))^2 \leq 1$$

$$-1 \leq \cos(x) \leq 1$$

$$0 \leq |\cos(x)| \leq 1$$

$$0 \leq (\cos(x))^2 \leq 1$$

$$-\frac{\pi}{2} < \tan^{-1}(x) < \frac{\pi}{2}$$

$$-1 < \tanh(x) < 1$$

$$0 < \operatorname{sech}(x) \leq 1$$